



A new Bayesian method for estimation of value at risk and conditional value at risk

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Abstract

Value at Risk (VaR) and Conditional Value at Risk (CVaR) have become the most popular measures of market risk in Financial and Insurance fields. However, the estimation of both risk measures is challenging, because it requires the knowledge of the tail of the distribution. Therefore, Extreme Value Theory initially seemed to be one of the best tools for this kind of problems, because using peaks-over-threshold method, we can assume the tail data approximately follow a Generalized Pareto distribution (GPD). The main objection to its use is that it only employs observations over the threshold, which are usually scarce. With the aim of improving the inference process, we propose a new Bayesian method that computes estimates built with all the information available. Informative prior Bayesian (IPB) method employs the existing relations between the parameters of the loss distribution and the parameters of the GPD that models the tail data to define informative priors in order to perform Metropolis–Hastings algorithm. We show how to apply IPB when the distribution of the observations is Exponential, stable or Gamma, to make inference and predictions. Afterwards, we perform a thorough simulation study to compare the accuracy and precision of the estimates computed by IPB and the most employed methods to esti-

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mate VaR and CVaR. Results show that IPB provides the most accurate, precise and least biased estimates, especially when there are very few tail data. Finally, data from two real examples are analysed to show the practical application of the method.

Keywords Risk measures · Generalized Pareto distribution · Bayesian inference · Informative priors

1 Introduction

Employing good measures of risk, that allow us to quantify uncertainty, is crucial in Risk Management. Value at Risk (VaR) and Conditional Value at Risk (CVaR) have become the most popular measures of market risk in Insurance and Finance, being also used in credit risk and operational risk issues. The applications of VaR and CVaR include reporting information, controlling and managing risk (Franke et al. 2004; Tsay 2005; Jorion 2007; Taniguchi et al. 2007; Jadhav and Ramanathan 2009; Embrechts et al. 2013). Their order properties have been widely studied in the literature (Jadhav and Ramanathan 2009).

However, the estimation of VaR and CVaR is challenging, because it requires the knowledge of the tail of the loss distribution. Historical or empirical method (Emp) is the simplest one, but it does not provide accurate estimates, especially when data are scarce. Bootstrap (Boot) can reduce mean squared error, but it usually offers biased estimates. Another option is to employ parametric methods, making classical inference (Par) or Bayesian inference (ParB). A review of methods can be found in Mina and Xiao (2001). As the objective is to make inference about the tail of the distribution, without the need to model its central part, the use of Extreme Value Theory seems to be natural (Gilli 2006; Bali 2007; Trzpiot and Majewska 2010; Park and Kim 2016; Van der Merwe et al. 2018; El Ghourabi et al. 2021; Kuang 2022). Peaks-over-threshold method is based on the property that the distribution of data above a fixed threshold u can be approximated by a properly scaled Generalized Pareto distribution, when u tends to the endpoint of the distribution (Balkema and De Haan 1974; Pickands 1975). Therefore, estimation of VaR and CVaR can be made considering that the observations over the threshold follow a GPD distribution.

Extreme Value Bayesian method (EVB) initially appeared to be a good choice, but unfortunately observations from the tail of the loss distribution are usually scarce, and then the method can lead to biased and not precise enough estimates. In this work, we propose a method to make estimation of VaR and CVaR, which not only seizes all the information the whole dataset provides, but also assigns tail data a higher weight in the inference process. In previous works, Martín et al. (2020, 2022) we already employed this idea to perform estimations for the parameters of Extreme Value distributions. The crucial issue is the use of the existing relations between the parameters of the loss distribution and the parameters of the GPD that models the tail of the observations. They are employed to define informative priors in order to perform Metropolis–Hastings (MH) algorithm. This procedure is called *informative prior Bayesian* (IPB) method.

We show how to apply IPB method when the distribution of data is Exponential, stable (focussing on Normal and Cauchy distributions) or Gamma, which are the most usual distributions in Risk Finance.

From a forecasting perspective, Engle and Manganelli (2004) and Gerlach et al. (2011) proposed new dynamic models for financial risk measures employing quantile regression methodology. Lu and Li (2011) introduced a new strategy using Monte Carlo for Value at Risk of portfolios modelling dependence through copulas, while Chen et al. (2012) and Gerlach et al. (2013) studied methods based on the asymmetric Laplace distribution. We also explain how to make predictions for VaR and CVaR employing the posterior predictive distribution computed through IPB method.

In Sect. 2, we introduce the definitions of VaR and CVaR and show their analytical expressions or how to compute them, when it is possible, for GPD, Gamma distribution and stable distributions, in particular for Cauchy and Normal distributions. In Sect. 3, we explain EVB and IPB method, detail how to perform them to make inference for the distributions considered and how to make predictions. Afterwards, in Sect. 4, we perform a thorough simulation study to compare the accuracy provided by all the inference methods for the distributions considered: Emp, Boot, Par, ParB, EVB and IPB. The results show that IPB provides the most accurate and precise estimates for VaR and CVaR. Finally, in Sect. 5, two real examples are studied to provide practical applications of IPB.

2 Risk measures

As we commented in the Introduction, VaR and CVaR are usually the employed measures to describe the tail of a loss distribution in a financial context.

Consider a continuous random variable X which represents the loss of an investment over a certain time horizon.

Definition 1 Given a parameter $0 < p < 1$, the Value at Risk (VaR_p) of X is the p -quantile of the distribution X

$$\text{VaR}_p(X) = \inf\{c : P(X \leq c) \geq p\} \quad (1)$$

and the Conditional Value at Risk (CVaR_p) of X is

$$\text{CVaR}_p(X) = E[X \mid X \geq \text{VaR}_p(X)] \quad (2)$$

VaR_p is the maximum loss expected in the $p \cdot 100\%$ of the best cases, while CVaR_p is the expected loss conditioned to the loss bigger than VaR_p . This measure is more sensitive to the shape of the tail of the loss distribution.

2.1 Generalized Pareto distribution

As there are several ways to parametrize GPD, in order to fix notation we will define that a random variable X is distributed as a GPD with shape parameter ξ and scale parameter σ , $X \sim \text{GPD}(\xi, \sigma)$ when its distribution function is

$$\text{GPD}(x; \xi, \sigma) = \begin{cases} 1 - \left(1 + \xi \frac{x}{\sigma}\right)^{-1/\xi} & \text{if } \xi \neq 0 \\ 1 - \exp\left(-\frac{x}{\sigma}\right) & \text{if } \xi = 0 \end{cases} \quad (3)$$

where $\xi \in \mathbb{R}$, $\sigma > 0$, with support $x \geq 0$ if $\xi \geq 0$ and $0 \leq x \leq -\frac{\sigma}{\xi}$ if $\xi < 0$.

By simple computations, given $0 < p < 1$, $\xi \neq 0$,

$$\text{VaR}_p(X) = \frac{\sigma}{\xi} \left[(1-p)^{-\xi} - 1 \right] \quad (4)$$

$$\text{CVaR}_p(X) = \frac{\sigma}{\xi} \left[\frac{1}{1-\xi} (1-p)^{-\xi} - 1 \right] \quad (5)$$

For $\xi = 0$, $\text{GPD}(0, \sigma)$ is the Exponential distribution with parameter $\lambda = \frac{1}{\sigma}$, and

$$\text{VaR}_p(X) = -\frac{1}{\lambda} \ln(1-p) \quad (6)$$

$$\text{CVaR}_p(X) = \frac{1}{\lambda} [1 - \ln(1-p)] \quad (7)$$

2.2 Gamma distribution

Gamma distribution is often employed to model financial data, so it is widely applied in the field of Risk Theory.

When $X \sim \text{Gamma}(\alpha, \beta)$, it is impossible to find an analytic expression for VaR_p , because there is not such expression for the distribution function. However, it is possible to deduce CVaR_p employing upper incomplete Gamma function $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$.

Given $0 < p < 1$,

$$\text{CVaR}_p(X) = \frac{1}{(1-p)\Gamma(\alpha)\beta} \Gamma(\alpha+1, \beta \text{VaR}_p(X)) \quad (8)$$

2.3 Stable distributions

Stable distributions are used to model physical and economic systems due to their important properties. Defined by Lévy (1925), they have been studied and applied in financial contexts, Embrechts et al. (2013), Nolan (2020). Besides, Yamai and Yoshihara (2002) showed different estimates for VaR and CVaR for stable distributions. By using the properties of stable distributions, the following result can be deduced.

Proposition 1 Let $Z \sim \text{Stable}(1, 0)$ and $X = aZ + b$, with parameters $a > 0$ and $b \in \mathbb{R}$. Given a parameter $0 < p < 1$,

$$\text{VaR}_p(X) = a\text{VaR}_p(Z) + b \quad (9)$$

$$\text{CVaR}_p(X) = a\text{CVaR}_p(Z) + b \quad (10)$$

By equations (9) and (10), VaR and CVaR could be computed for the rest of stable distributions, assuming we know VaR and CVaR for the corresponding standard Z distribution. In this work, we show how to compute VaR and CVaR for Normal and Cauchy distributions, because they are the most popular and employed stable distributions.

Corollary 2 Let $X \sim \mathcal{N}(\mu, \sigma)$ be a random variable with mean $\mu \in \mathbb{R}$ and standard deviation $\sigma > 0$. Given a parameter $0 < p < 1$,

$$\text{VaR}_p(X) = \mu + \sigma z_p \quad (11)$$

$$\text{CVaR}_p(X) = \mu + \frac{\sigma}{(1-p)\sqrt{2\pi}} \exp\left\{-\frac{1}{2}z_p^2\right\} \quad (12)$$

where z_p is the p -quantile of $Z \sim \mathcal{N}(0, 1)$.

Cauchy distribution is a stable distribution with heavy tails, and it has the special characteristic that it does not present finite moments, so it is not possible to determine its mean. Moreover, it is impossible to define the risk measure CVaR. However, unlike Normal distribution, this time the risk measure VaR can be computed theoretically.

Corollary 3 Let $X \sim \mathcal{C}(\gamma, \delta)$ be a random variable with location parameter $\gamma \in \mathbb{R}$ and scale parameter $\delta > 0$. Given a parameter $0 < p < 1$,

$$\text{VaR}_p(X) = \gamma + \delta \tan\left[\pi\left(p - \frac{1}{2}\right)\right] \quad (13)$$

3 Risk measures estimation

As we commented in the Introduction, several methods have been employed to make estimation of VaR and CVaR: historical or empirical method (Emp), Bootstrap (Boot), parametric classical inference (Par) or parametric Bayesian inference (ParB).

EVT provides tools to estimate tail data. Therefore, it seems reasonable to employ EVT to compute estimates for VaR and CVaR. Peaks-over-threshold (POT) method is based on the property that the distribution of data above a fixed threshold u can be approximated by a properly scaled Generalized Pareto distribution, when u tends to the endpoint of the distribution, Balkema and De Haan (1974); Pickands (1975).

Let X be a random variable and u the threshold value. The random variable $X_u = X - u \mid X > u$ represents the exceedances over u . Then, for a sufficiently high threshold u , the random variable X_u follows a scaled Generalized Pareto distribution, $\text{GPD}(\xi, \sigma)$.

Regarding to the threshold choice, in financial contexts, the most usual way to define it is by fixing the probability below the threshold p_u , that is, by choosing the p_u -quantile of X . The following result allows us to compute risk measures for the loss distribution X through the risk measures for the GPD that models the tail, once the threshold u is fixed.

Proposition 4 *Let u be the threshold defined as the p_u -quantile of X . Given $0 < p < 1$ that satisfies $p > p_u$, VaR_p and CVaR_p of X can be computed this way*

$$\text{VaR}_p(X) = u + \text{VaR}_{p_t}(X_u) \quad (14)$$

$$\text{CVaR}_p(X) = u + \text{CVaR}_{p_t}(X_u) \quad (15)$$

where $p_t = 1 - \frac{1-p}{1-p_u}$ and $\text{VaR}_{p_t}(X_u)$, $\text{CVaR}_{p_t}(X_u)$ are computed using Eqs. (4–7).

We will explain below how to make use of Proposition 4 for constructing a Bayesian method based on EVT, that we will call Extreme Value Bayesian method (EVB), performing MH algorithm.

Let $\mathbf{x} = (x^1, \dots, x^n)$ be a sample of size n of the random variable X with parameters θ and let $\mathbf{x}_u = (x_u^1, \dots, x_u^m)$ be the sample of size m of the random variable X_u .

3.1 Extreme value Bayesian method

We will consider non-informative prior distributions to estimate the parameters ξ and σ of the GPD using tail values (details of these estimates can be consulted in Martín et al. 2022) and then apply the results of Proposition 4 to compute estimates for VaR and CVaR.

The steps of this method are:

Do $j = 1$, start parameters $\xi^{(0)}$, $\sigma^{(0)}$

Generate candidate parameters

$$\xi^* \sim \mathcal{N}(\xi^{(j-1)}, v_\xi), \quad \sigma^* \sim \mathcal{N}(\sigma^{(j-1)}, v_\sigma)$$

Evaluate the probability of acceptance $\tau = \min\{1, r_{\xi, \sigma}\}$, where the ratio is

$$r_{\xi, \sigma} = \frac{\pi(\xi^*, \sigma^* | \mathbf{x}_u, \xi^{(j-1)}, \sigma^{(j-1)})}{\pi(\xi^{(j-1)}, \sigma^{(j-1)} | \mathbf{x}_u, \xi^*, \sigma^*)}$$

Generate $u \sim \mathcal{U}(0, 1)$.

If $u < \tau$, do $\xi^{(j)} = \xi^*$ and $\sigma^{(j)} = \sigma^*$, else do $\xi^{(j)} = \xi^{(j-1)}$ and $\sigma^{(j)} = \sigma^{(j-1)}$

Do $j = j + 1$

However, as we mentioned before, since this method only employs the observations above the threshold, which are usually scarce, it may provide biased estimates. Then,

with the aim of improving inference process, but keeping the spirit of EVT, we will introduce informative prior Bayesian method.

3.2 Informative prior Bayesian method

Informative Prior Bayesian method includes informative prior distributions for the parameters of the GPD, obtained from the relationship between the parameters θ of the loss distribution X and the GPD that models the tail. We compute estimates for VaR and CVaR for the exceedances over the threshold and by Proposition 4, the estimates for risk measures of X can also be calculated. Consequently, we incorporate all the information available for the whole set of observations to the procedure, but also give more weight to tail data.

Let $g_\xi(\theta)$, $g_\sigma(\theta)$ be the functions that define the relationships between ξ and σ , respectively, and θ . The steps of this method are:

Do $j = 1$, start parameters $\xi^{(0)}$, $\sigma^{(0)}$
 Generate a parameter $\theta^{(j)}$ employing the posterior distribution $\pi(\theta | \mathbf{x})$
 Generate candidate parameters

$$\xi^* \sim \mathcal{N}\left(g_\xi\left(\theta^{(j)}\right), v_\xi\right), \quad \sigma^* \sim \mathcal{N}\left(g_\sigma\left(\theta^{(j)}\right), v_\sigma\right)$$

Evaluate the probability of acceptance $\tau = \min\{1, r_{\xi, \sigma}\}$,
 where

$$r_{\xi, \sigma} = \frac{\pi\left(\xi^*, \sigma^* | \mathbf{x}_u, \xi^{(j-1)}, \sigma^{(j-1)}\right)}{\pi\left(\xi^{(j-1)}, \sigma^{(j-1)} | \mathbf{x}_u, \xi^*, \sigma^*\right)}$$

Generate $u \sim \mathcal{U}(0, 1)$.

If $u < \tau$, do $\xi^{(j)} = \xi^*$ and $\sigma^{(j)} = \sigma^*$, else do $\xi^{(j)} = \xi^{(j-1)}$ and $\sigma^{(j)} = \sigma^{(j-1)}$

Do $j = j + 1$

Next, this method is developed for the most employed loss distributions.

3.2.1 Exponential distribution

When $X \sim \text{Exp}(\lambda)$, $X_u \sim \text{Exp}(\lambda) = \text{GPD}(0, 1/\lambda)$ and $g_\sigma(\lambda) = 1/\lambda$. Given that the prior distribution is symmetric, the probability of acceptance of the MH algorithm can easily be computed using the likelihood functions, therefore

$$r_\sigma = \left(\frac{\sigma^{(j-1)}}{\sigma^*}\right)^m \exp\left\{\left(\frac{1}{\sigma^{(j-1)}} - \frac{1}{\sigma^*}\right) \sum_{i=1}^m x_u^i\right\} \quad (16)$$

Table 1 Estimates for GPD's parameters

Distribution	ξ_Z	σ_Z
$\mathcal{N}(0, 1)$	$-0.7 + 0.61p_u$	$0.34 + 3.18(1 - p_u) - 12.4(1 - p_u)^2$
$C(0, 1)$	1	$\frac{1}{\pi}(1 - p_u)^{-1}$

3.2.2 Stable distributions

Let $Z \sim \text{Stable}(1, 0)$ and $X = aZ + b$. For a sufficiently high threshold u , it is satisfied that $Z_u \sim \text{GPD}(\xi_Z, \sigma_Z)$, then consequently $X_u \sim \text{GPD}(\xi_Z, a \cdot \sigma_Z)$. Therefore, it is enough to find estimates for the parameters ξ_Z and σ_Z . In Martín et al. (2022), a simulation study was carried out in order to find estimates of standard stable distributions Normal and Cauchy, for a threshold u defined as the p_u -quantile. Table 1 shows the estimates found in that work.

Therefore, to estimate the GPD parameters ξ and σ , we can just consider the prior distributions

$$\xi \sim \mathcal{N}(\xi_Z, v_\xi), \quad \sigma \sim \mathcal{N}(a \cdot \sigma_Z, v_\sigma)$$

The ratios of MH algorithm for each parameter are

$$r_\xi = \exp \left\{ \frac{1}{2v_\xi^2} \left((\xi^{(j-1)} - \xi_Z)^2 - (\xi^* - \xi_Z)^2 \right) - \left(1 + \frac{1}{\xi^*} \right) \sum_{i=1}^m \ln \left(1 + \xi^* \frac{x_u^i}{\sigma^{(j-1)}} \right) + \left(1 + \frac{1}{\xi^{(j-1)}} \right) \sum_{i=1}^m \ln \left(1 + \xi^{(j-1)} \frac{x_u^i}{\sigma^{(j-1)}} \right) \right\} \quad (17)$$

$$r_\sigma = \left(\frac{\sigma^{(j-1)}}{\sigma^*} \right)^m \exp \left\{ \frac{1}{2v_\sigma^2} \left((\sigma^{(j-1)} - a \cdot \sigma_Z)^2 - (\sigma^* - a \cdot \sigma_Z)^2 \right) - \left(1 + \frac{1}{\xi^{(j-1)}} \right) \sum_{i=1}^m \ln \left(1 + \xi^{(j-1)} \frac{x_u^i}{\sigma^*} \right) + \left(1 + \frac{1}{\xi^{(j-1)}} \right) \sum_{i=1}^m \ln \left(1 + \xi^{(j-1)} \frac{x_u^i}{\sigma^{(j-1)}} \right) \right\} \quad (18)$$

3.2.3 Gamma distribution

When $X \sim \text{Gamma}(\alpha, \beta)$, a simulation study has been carried out, similar to that of stable distributions, with the aim of obtaining empirical relationships between its parameters and GPD parameters ξ, σ . Resulting relationships are

$$g_\xi(\alpha, \beta) = -0.032 + 0.014\alpha^{-1}, \quad g_\sigma(\alpha, \beta) = \frac{1}{2}\beta^{-1}(1 + \sqrt{\alpha}) \quad (19)$$

These empirical relationships allow for the construction of informative prior distributions for ξ and σ . Therefore,

$$\xi \sim \mathcal{N}(g_{\xi}(\alpha, \beta), v_{\xi}), \quad \sigma \sim \mathcal{N}(g_{\sigma}(\alpha, \beta), v_{\sigma})$$

The ratio is given by the following expression:

$$r_{\xi, \sigma} = \left(\frac{\sigma^{(j-1)}}{\sigma^*} \right)^m \cdot \prod_{i=1}^m \left[\left(1 + \xi^* \frac{x_u^i}{\sigma^*} \right)^{-(1+1/\xi^*)} \cdot \left(1 + \xi^{(j-1)} \frac{x_u^i}{\sigma^{(j-1)}} \right)^{1+1/\xi^{(j-1)}} \right] \quad (20)$$

3.2.4 Prediction

As discussed in the Introduction, an interesting issue about risk measures is how to make predictions for them. In a Bayesian framework, the posterior predictive distribution is employed to forecast values. This is,

$$\pi(x | \theta) = \int_{\theta \in \Theta} f(x; \theta) \cdot \pi(\theta | \mathbf{x}) d\theta, \quad (21)$$

where $\theta = (\xi, \sigma)$, $\pi(\theta | \mathbf{x})$ is the posterior distribution, and $f(x; \theta)$ is the density function of the GPD distribution.

The steps to predict with this method are:

Do $j = 1$, start parameters $\xi^{(0)}, \sigma^{(0)}$

Generate $\xi^{(j)}, \sigma^{(j)}$

Generate $x_{pred}^{(j)}$ with GPD($x; \xi^{(j)}, \sigma^{(j)}$) model

Do $j = j + 1$

The obtained $\mathbf{x}_{pred}$ allows us to forecast risk measures for a GPD distribution. The p_t -quantile of $\mathbf{x}_{pred}$ is a prediction of VaR_{p_t} and the mean of the values that exceed VaR_{p_t} is a prediction of CVaR_{p_t} , following the notation of Proposition 4. Finally, predictions of the risk measures VaR_p and CVaR_p for the loss distribution can be computed.

4 Results

In this Section, we show the results of a thorough simulation study in order to compare the quality of the estimates for VaR and CVaR computed with the different inference methods: Empirical (Emp), Bootstrap (Boot), parametric classical (Par), Bayesian Parametric (ParB), Extreme Value Bayesian (EVB) and informative prior Bayesian (IPB); for the distributions cited in previous sections: Exponential, Normal, Cauchy and Gamma.

The value of the threshold must be high enough so that the GPD is a good approximation for X_u distribution, but small enough to define a reasonably high number of tail data m . A reasonable value would be the quantile of order 0.9. In order to study

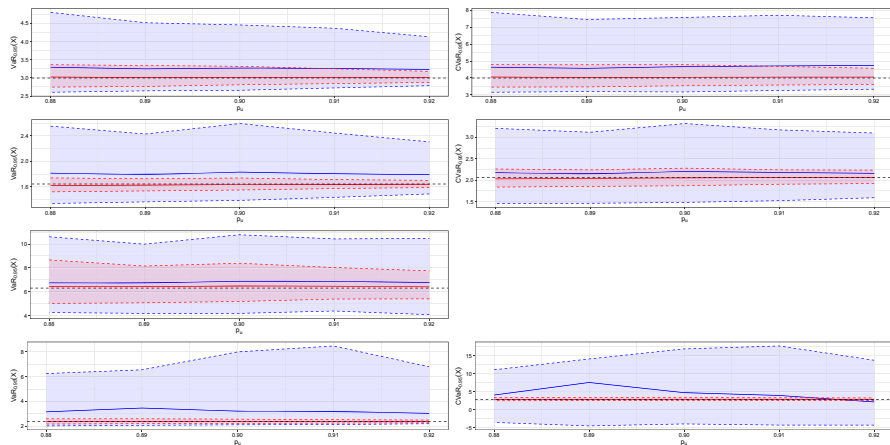


Fig. 1 Estimates of $\text{VaR}_{0.95}(X)$ (left charts) and $\text{CVaR}_{0.95}(X)$ (right charts), for $X \sim \text{Exp}(1)$ (upper charts), $X \sim \mathcal{N}(0, 1)$ (second rows charts), $X \sim \mathcal{C}(0, 1)$ (third rows charts) and $X \sim \text{Gamma}(2, 2)$ (lower charts) vs different values of order p_u of the threshold u , with 2.5% and 97.5% probability bounds, by EVB (blue lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

whether the choice of the threshold affects the quality of the estimates of the risk measures, $\text{VaR}_p(X)$ and $\text{CVaR}_p(X)$, we chose different values of the threshold, which was defined as the p_u -quantile of X . Concretely, p_u took the values between 0.88 and 0.92, by increments of length 0.01. We obtained a Monte Carlo Markov chain (MCMC) of length 10,000, taking 3000 training values and a value of 50 for the thinning. Finally, each sequence was repeated 500 times, and we computed the mean to estimate VaR or CVaR for distributions $\text{Exp}(1)$, $\mathcal{N}(0, 1)$, $\mathcal{C}(0, 1)$ and $\text{Gamma}(2, 2)$, with samples of size $n = 32$.

Figure 1 shows that IPB offers stable results for the estimates of $\text{VaR}_{0.95}(X)$ and $\text{CVaR}_{0.95}(X)$, with respect to the value of p_u . Estimates computed by EVB offer much wider probability bounds and biased estimates. Anyway, we can conclude that the choice of the threshold does not affect the estimates of the risk measures for the methods, as the slope of most estimates is nearly horizontal. Therefore, we will fix the value of the threshold for the rest of the simulations as the quantile of 0.9 order, $p_u = 0.9$.

Afterwards, once the threshold was fixed, we studied the results of the estimates for different values of their order. Figures 2, 3 show the error distributions of the risk measures for three of these values, $p = 0.95, 0.97$ and 0.99 , and for all the methods considered. It can be observed that all the methods show a similar qualitative behaviour, although the bigger the value of p , the worse the results get, both in accuracy and precision. However, goodness order between them keeps stable. Notice that ParB, and especially IPB, provide more concentrated and less biased estimates, regardless of the distribution and value of p considered. Although Par also provides accurate estimates, it exhibits a larger dispersion than ParB and IPB.

Consequently, we compared ParB and IPB in more detail for Exponential, stable (Normal and Cauchy) and Gamma distributions, for different parameter values and sample sizes. For each distribution and fixed parameters, samples of sizes $n = 2^i$, $i = 5, \dots, 10$ were simulated. So, we obtained a Monte Carlo Markov chain (MCMC) of

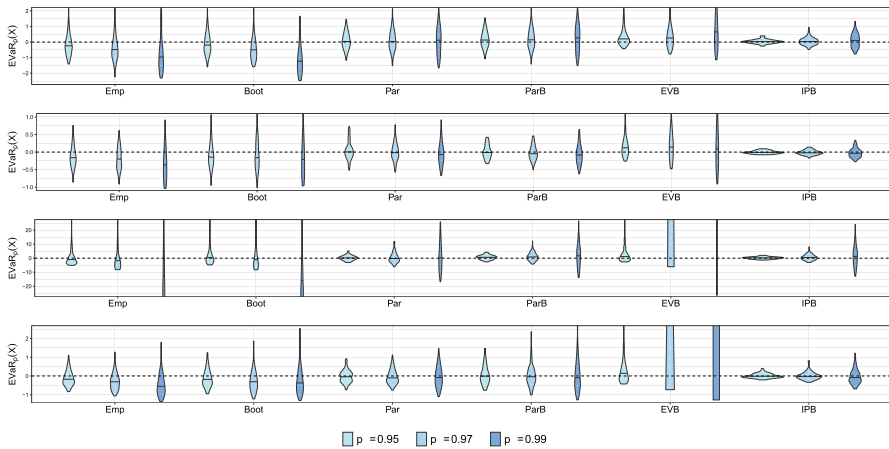


Fig. 2 Error distributions of $VaR_p(X)$ for $X \sim \text{Exp}(1)$ (upper row), $X \sim \mathcal{N}(0, 1)$ (second row), $X \sim C(0, 1)$ (third row) and $X \sim \text{Gamma}(2, 2)$ (lower row)

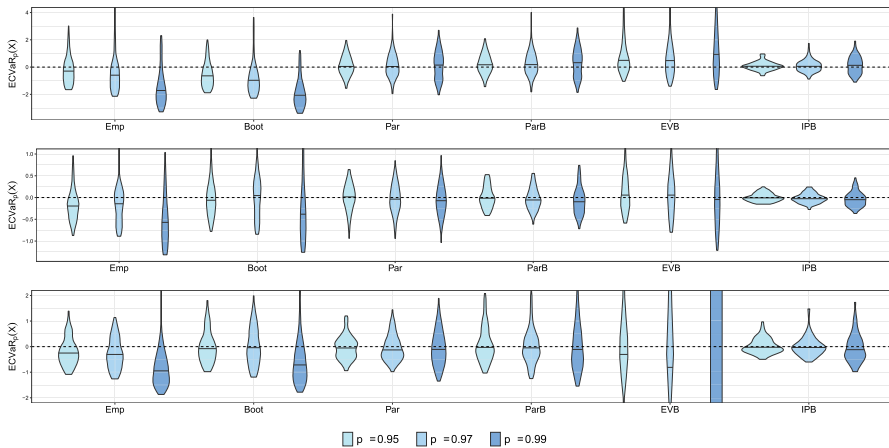


Fig. 3 Error distributions of $CVaR_p(X)$ for $X \sim \text{Exp}(1)$ (upper row), $X \sim \mathcal{N}(0, 1)$ (middle row) and $X \sim \text{Gamma}(2, 2)$ (lower row)

length 10000, taking 3000 training values and a value of 50 for the thinning. Each sequence was repeated 100 times, and we computed the mean to estimate VaR or CVaR. Also, we calculated the quantiles of 0.025 and 0.975 orders to determine the probability bounds for each risk measure.

4.1 Exponential distribution

Let $X \sim \text{Exp}(\lambda)$. We considered $\lambda = 2^j$, $j = -2, -1, 0, 1, 2$. Figure 4 shows the estimates for $VaR_{0.95}(X)$ (upper row) and $CVaR_{0.95}(X)$ (lower row), along with 2.5% and 97.5% probability bounds, computed with ParB (green lines) and IPB (red lines) for $\lambda = 0.5$ (left charts), $\lambda = 1$ (centre charts) and $\lambda = 2$ (right charts). Besides, the real values of each risk measure are drawn as a black dashed line.

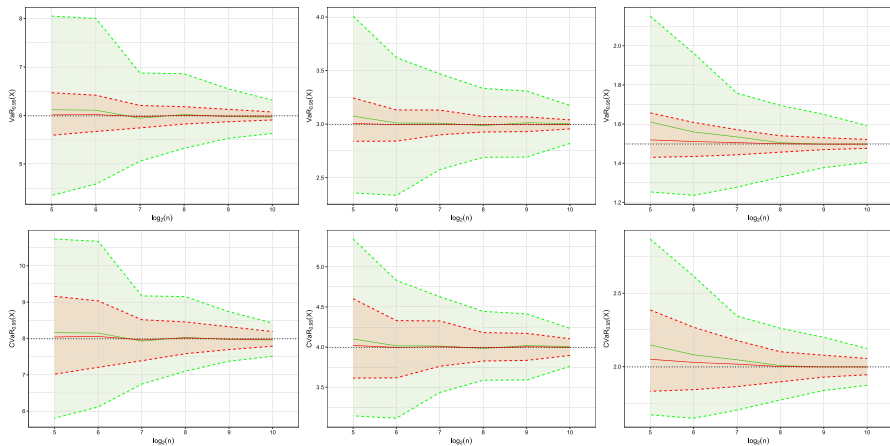


Fig. 4 Estimates of $\text{VaR}_{0.95}(X)$ (upper row) and $\text{CVaR}_{0.95}(X)$ (lower row), for $X \sim \text{Exp}(\lambda)$ with $\lambda = 0.5$ (left charts), $\lambda = 1$ (centre charts) and $\lambda = 2$ (right charts), with 2.5% and 97.5% probability bounds, by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

As we can see, ParB offers worse estimates, especially when data are scarce, while IPB offers more accurate and precise estimates for both risk measures for any size of the dataset n . We also observe that the deviation obtained for small values of n is larger for the ParB method, and it can be best appreciated when the value of λ is bigger. Again, estimates from IPB remain stable.

4.2 Stable distributions

Let $X \sim \text{Stable}(a, b)$. For stable distributions, the location parameter b does not have an influence on the estimates of the parameters of the GPD; therefore, it does not affect the estimates of the risk measures. So, we fixed $b = 0$. The scale parameter a does affect the estimates; therefore, we took $a = 2^j$, $j = -2, -1, 0, 1, 2$. The most representative cases $a = 0.5, 1, 2$ are shown in Figs. 5, 6. In particular, Fig. 5 shows the estimates of VaR and CVaR for Normal distribution and Fig. 6 shows the estimates for VaR for Cauchy distribution. In both figures, we can see how ParB (green lines) provides worse estimates, especially when there is a lack of data. In contrast, IPB (red lines) provides accurate estimates for any size n , for any value of the parameter.

In particular, for the Cauchy distribution, when the size n is low, ParB offers estimates of VaR with really high deviations from the real value, especially when the scale parameter is high.

4.3 Gamma distribution

Let $X \sim \text{Gamma}(\alpha, \beta)$. We considered $\alpha = 2^k$, $k = -2, 1, 1, 2$ and $\beta = 2^j$, $j = -2, -1, 0, 1, 2$. Figures 7, 8 show the estimates of $\text{VaR}_{0.95}(X)$ and $\text{CVaR}_{0.95}(X)$ for different values of $\beta = 0.25, 1$ and 4, for $\alpha = 0.5$ (Fig. 7) and $\alpha = 2$ (Fig. 8).

As we can see, charts from Fig. 7 are quite similar in shape to the ones from Fig. 8. As in the previous cases, ParB (green lines) offers worse estimates, especially when

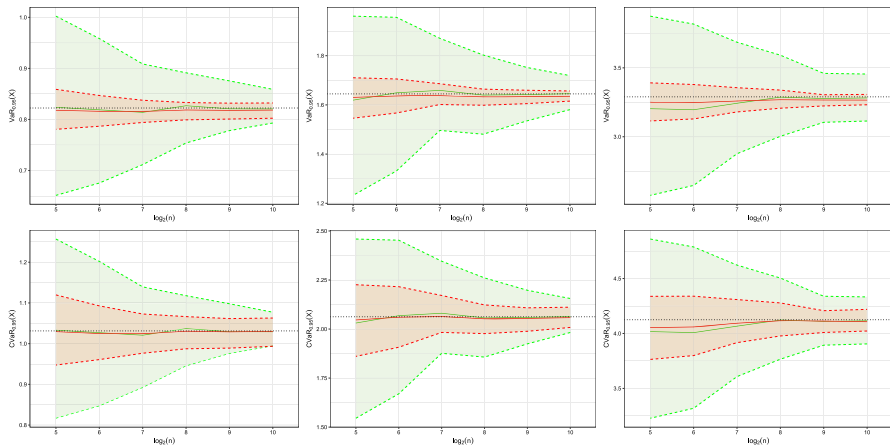


Fig. 5 Estimates of $VaR_{0.95}(X)$ (upper charts) and $CVaR_{0.95}(X)$ (lower charts), for $X \sim \mathcal{N}(0, a)$ with $a = 0.5$ (left charts), $a = 1$ (centre charts) and $a = 2$ (right charts), with 2.5% and 97.5% probability bounds, by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

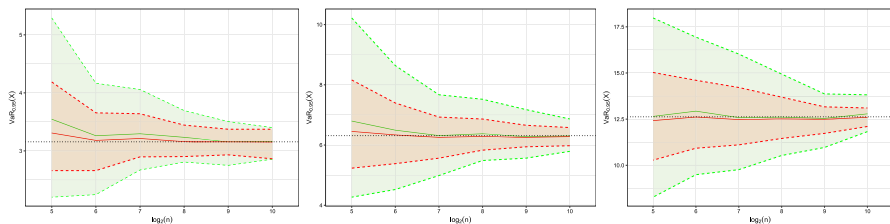


Fig. 6 Estimates of $VaR_{0.95}(X)$ for $X \sim \mathcal{C}(0, a)$ with $a = 0.5$ (left chart), $a = 1$ (centre chart) and $a = 2$ (right chart), with 2.5% and 97.5% probability bounds, by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

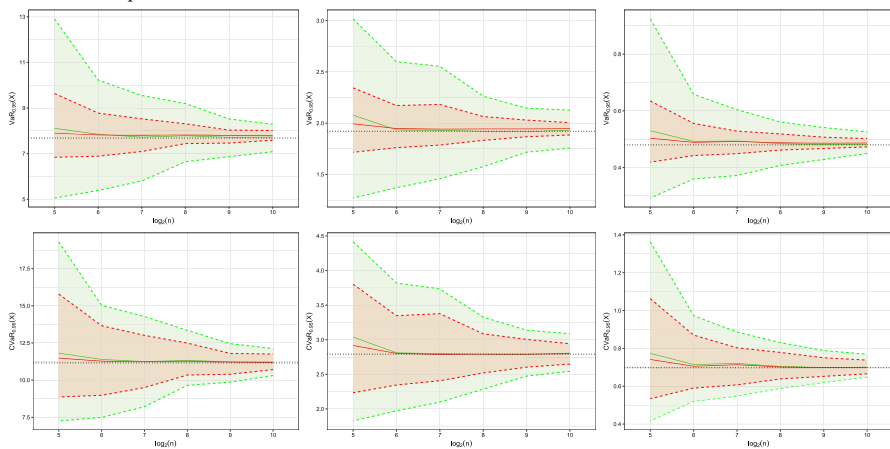


Fig. 7 Estimates of $VaR_{0.95}(X)$ (upper row) and $CVaR_{0.95}(X)$ (lower row), for $X \sim \text{Gamma}(0.5, \beta)$ with $\beta = 0.25$ (left charts), $\beta = 1$ (centre charts) and $\beta = 4$ (right charts), with 2.5% and 97.5% probability bounds, by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

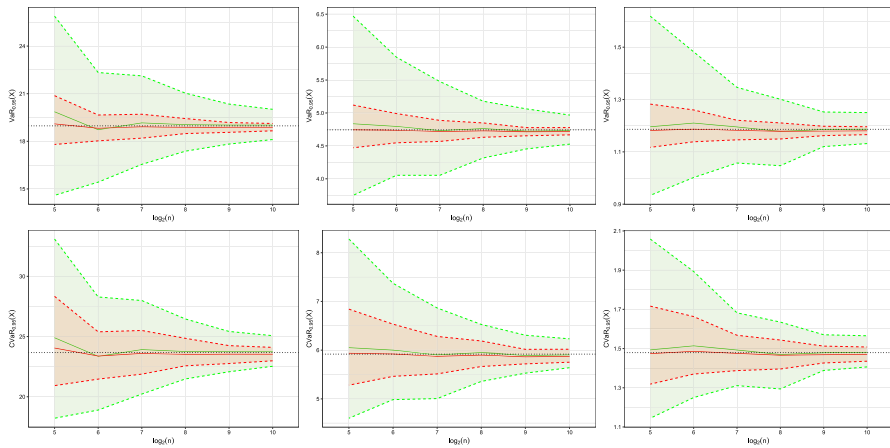


Fig. 8 Estimates of $\text{VaR}_{0.95}(X)$ (upper row) and $\text{CVaR}_{0.95}(X)$ (lower row), for $X \sim \text{Gamma}(2, \beta)$ with $\beta = 0.25$ (left charts), $\beta = 1$ (centre charts) and $\beta = 4$ (right charts), with 2.5% and 97.5% probability bounds, ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure

the number of data n is low and for low values of β . IPB (red lines) provides very accurate estimates of both measures, regardless of the value of the parameters and the size of the sample.

5 Applications

In this Section, we will show two practical applications of IPB method in order to compute VaR and CVaR. Usually, in Financial Risk studies, the variable of interest to compute risk measures is the daily logarithmic return of stock index, defined by

$$R_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (22)$$

where P_t and P_{t-1} indicate the stock index on day t and $t - 1$, respectively.

Usually, a quantity of interest is the maximum loss expected in the 95% of the worst cases, that is, $\text{VaR}_{0.05}$.

Typically, risk measures of R_t are computed through time series models employing historical data from $R_1, R_2, \dots, R_{t-1}$ (Likitratcharoen et al. 2021, 2023). However, these models need a large number of data to obtain precise values, because there are few extreme values.

To show the advantage of IPB, a study was carried out to compare it with ParB.

5.1 Standard and Poor's 500 index dataset

In the USA, the Standard and Poor's 500 index, S&P500, is a stock market index based on the market capitalization of 500 large companies with shares listed on the US stock exchange.

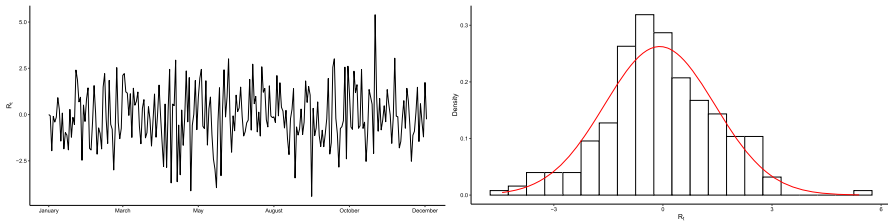


Fig. 9 S&P500's daily logarithmic return (%) (left chart) for 2022 and its histogram(right chart). Red curve represents the density function of $\mathcal{N}(-0.09, 1.52)$

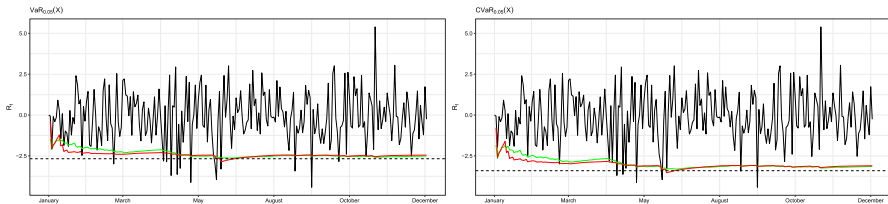


Fig. 10 Estimates of $\text{VaR}_{0.05}(R_t)$ (left chart) and $\text{CVaR}_{0.05}(R_t)$ (right chart) by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure. Black curves represent the S&P500 daily logarithmic return (%)

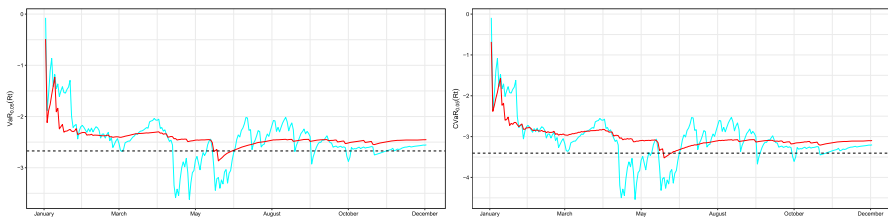


Fig. 11 Predictions of $\text{VaR}_{0.05}(R_t)$ (left chart) and $\text{CVaR}_{0.05}(R_t)$ (right chart) by IPB (red lines) and GARCH(1,1) model (cyan lines). Black dashed lines represent the real value of each risk measure

For this study, data of S&P500's daily closing for year 2022 were selected (SP500 2023). Figure 9 (left chart) shows R_t (in percentage) for 2022. The dataset comprehends a sample of 251 values, most of them between -4.0% and 4.0% .

Figure 9 (right chart) shows the behaviour of the data, and it can be observed that data can be fitted to a normal distribution. Therefore, we will consider that quantile $\text{VaR}_{0.05}$ is equal to $-\text{VaR}_{0.95}$, so we shall employ the negative values of R_t for the computations.

Figure 10 shows the estimates for VaR and CVaR provided by ParB (green lines) and IPB (red lines) employing historical data. IPB provides better estimates than ParB when data is scarce. When the time increases, there are more data to make estimations and both methods provide estimates of similar quality.

Generally, GARCH(1,1) temporal models are used to predict financial risk measures. However, IPB also allows for prediction of these measures, as we explained in Sect. 3.2.4. Figure 11 shows the predictions provided by IPB (red line) and Garch(1,1) model (cyan line). It can be observed that IPB provides more stable predictions.

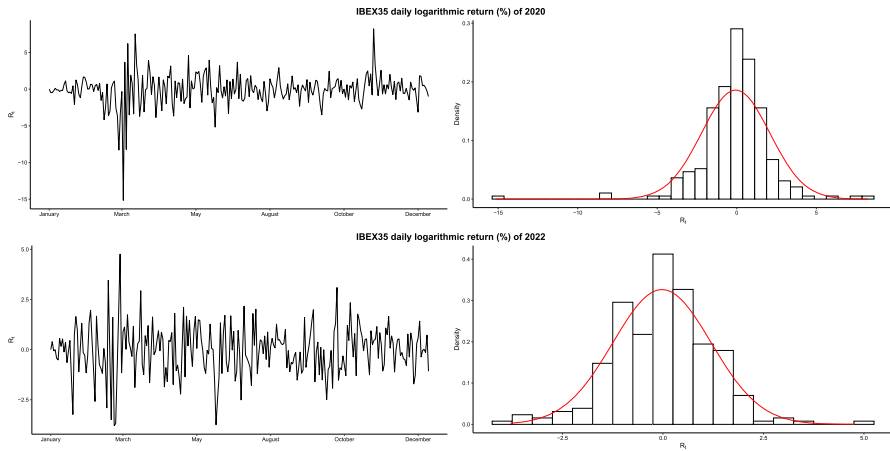


Fig. 12 IBEX35's daily logarithmic return (%) (left charts) and its histogram (right charts) for 2020 (upper charts) and 2022 (lower charts). Red curves represent the density function of $\mathcal{N}(-0.07, 2.15)$ (upper) and $\mathcal{N}(-0.02, 1.22)$ (lower)

5.2 Iberia index dataset

In Spain, the most important stock index is Iberia Index or IBEX35. This index is equivalent to S&P500 index. For this study, the data of IBEX35's daily closing for year 2022 were selected (IBEX35 2023). Besides, in order to study the possible consequences of the COVID-19 pandemic on this stock index, data for year 2020 were also taken.

Figure 12 (left charts) shows R_t (in percentage) for 2020 and 2022, respectively. In both cases, the majority of these values range between -5.0% and 5.0% , but it should be noted that in March of 2020, R_t took a value of -15% due to COVID-19 pandemic. For both years, the dataset comprehends a sample of 257 values. Also, it can be observed that data fit a Normal distribution.

Figure 13 shows the estimates for VaR and CVaR provided by ParB (green lines) and IPB (red lines) employing historical data. Again, IPB provides better estimates than ParB.

Figure 14 shows the predictions for risk measures by IPB (red lines) and GARCH(1,1) model (cyan lines). Again, IPB provides stable predictions when the time increases.

6 Conclusions

1. In this work, we proposed a new Bayesian method to estimate VaR and CVaR, which are the most popular risk measures. This method, Informative Prior Bayesian, is based on introducing informative priors by using the relation between the parameters of the loss distribution and the tail GPD parameters. Therefore, we employ EVT tools, but also incorporate information from the whole dataset

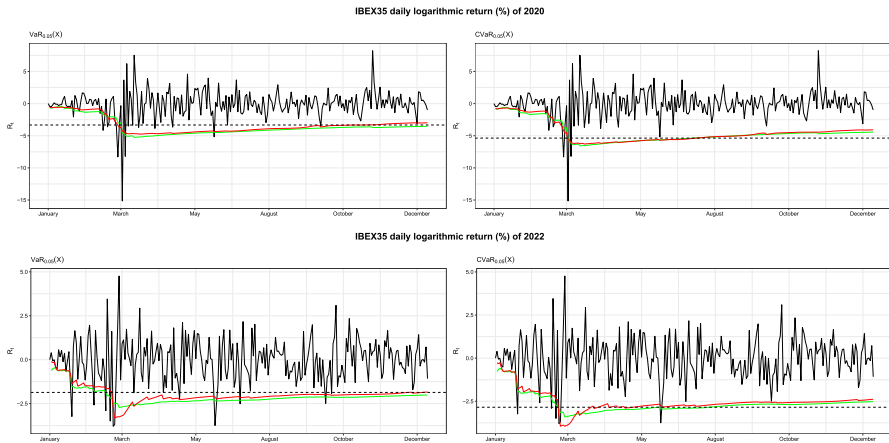


Fig. 13 Estimates of $VaR_{0.05}(R_t)$ (left charts) and $CVaR_{0.05}(R_t)$ (right charts) by ParB (green lines) and IPB (red lines). Black dashed lines represent the real value of each risk measure. Black curves represent the IBEX35 daily logarithmic return (%)

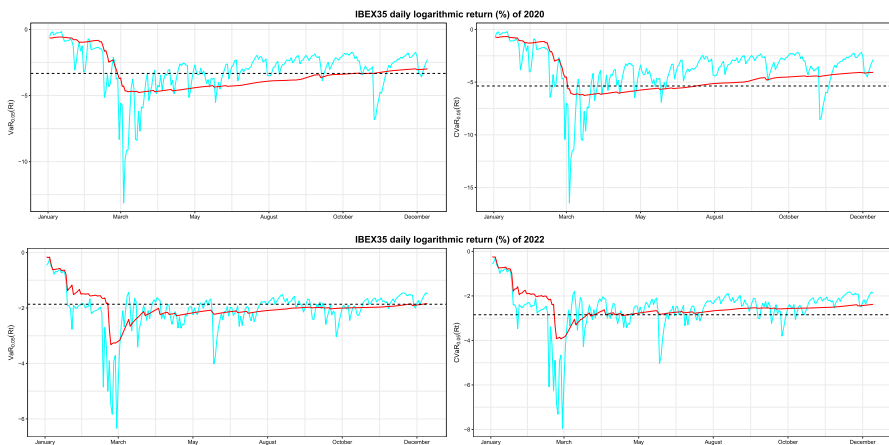


Fig. 14 Predictions of $VaR_{0.05}(R_t)$ (left charts) and $CVaR_{0.05}(R_t)$ (right charts) by IPB (red lines) and GARCH(1,1) model (cyan lines). Black dashed lines represent the real value of each risk measure

of observations. We compared the performance of Informative Prior Bayesian with other usual estimation methods for VaR and CVaR: Empirical, Bootstrap, parametric classical, Bayesian Parametric and a simple Extreme Value Bayesian method.

2. We focussed our study on Exponential, Normal, Cauchy and Gamma distributions, which are the most employed loss distributions.
3. For each distribution, we carried out a simulation study to check whether the choice of the threshold and the order p of VaR and CVaR affect the quality of the estimates given by the different estimation methods, concluding that neither the threshold nor the order affect the goodness the inference process. Besides,

Bayesian Parametric and especially Informative Prior Bayesian are selected as the best estimation methods.

4. Fixing the threshold and p , we performed a wide simulation study for all the distributions considered, to compare the accuracy and precision of the estimates for VaR and CVaR given by Informative Prior Bayesian and Bayesian Parametric. For any distribution or chosen parameter, Informative Prior Bayesian provided more accurate, less biased and more precise estimates, especially when the dataset was scarce.
5. In real situations, distribution of data is generally unknown. We considered two examples with real data and showed how Informative Prior Bayesian is more appropriate than Bayesian Parametric to perform estimations, having a more stable behaviour and lower dispersion.
6. Also, Informative Prior Bayesian allows for prediction of the risk measures studied. We performed forecasting for VaR and CVaR on the examples to compare their quality against the usual GARCH(1,1) model. We concluded that Informative Prior Bayesian provides more stable predictions than GARCH model.

Author Contributions These authors contributed equally to this work.

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Declarations

Conflict of interest The code is available at <http://rpubs.com/MaribelPA/1214204>.

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